

Ben Alvaro

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Education

The University of Melbourne (WAM: 83)

Bachelor of Science (Computing and Software Systems)

March 2025 – November 2027 (Expected)

Melbourne, Vic

Merewether High School

HSC (ATAR 97.2)

January 2019 – September 2024

Newcastle, NSW

Relevant Subjects

Design of Algorithms, Database Systems, Elements of Data Processing, Object Oriented Programming

Experience

Student ML Engineer

Pitcrew AI

December 2023 – Present

Newcastle, NSW (Remote)

- Contributed to thermal imaging ML model training and troubleshooting including calibration and installation of hardware, including for clients based internationally
- Constructed and executed a model-training pipeline that resulted in a successful model pushed to production
- Utilised Jira, Confluence, CVAT and Slack to effectively supervise annotation process

Lead Software Engineer

Melbourne University Racing / University of Melbourne

March 2026 – Present

Melbourne, VIC

- Led a team of 7 to design, develop and integrate a working firmware and telemetry service for the M26, a Formula SAE Car
- Managed the team wide project to meet tight deadlines, working with and around other sub teams to integrate the system with other parts of the car
- Extended out of my comfort zone, learning new skills during manufacturing, including soldering, machining and high voltage safety

IT Director

UniMelb CISSA

March 2025 – Present

Melbourne, VIC

- Released new club website (cissa.org.au) and built internal infrastructure
- Created interactive pages and implemented Website plug-ins using Python, react, and Node.JS
- Worked within the IT Team and between subcommittees to plan and run events totalling 1000+ participants across the year, and deployed internal tools to organise finances

Projects

M26 Formula SAE Car | *C, Zephyr, Python, Embedded Systems*

March - December 2026

- Led a team of 7 to develop from the ground up the software for an electric race car, to compete at Formula SAE-A in Tailem Bend, South Australia
- Components involved embedded systems, wireless telemetry for data, and integrating CAN buses on the car

Machine Learning-based Game AI | *Python, PyTorch*

June 2025

- Developed a variety of AIs for different games such as Snake & Space Invaders using reinforcement learning
- Implemented using PyTorch for ML process, PyGame for games and other tools for data visualization

F1 Analytics Platform | *Python, Matplotlib, Pandas*

April 2026

- Developed a functional platform to analyse how laptimes change over time in relation to other factors
- Utilised Unsupervised ML techniques & algorithms such as anomaly detection and association rule to find different influences to lap times, and isolate these effects

Skills and Proficiencies

Languages: Python, C/C++, HTML/CSS, JavaScript, Java, SQL

Developer Tools: VS Code, Visual Studio, XCode, Git, Confluence, CVAT, Jira, Pandas

Soft Skills: Communication, Problem Solving, Project Management, Task Prioritization, Conflict Resolution

Certificates

AWS Certified AI Practitioner, CompTIA A+ (220-1101), CCNA 200-301 (Ongoing)